UNIT 5 · 8 hours

Introduction to Compensators



Compensators, Controllers and State-Space Analysis

Control Systems (22EEG53)
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Learning objectives

- Explain why compensation is needed
- Describe series, feedback and feed-forward compensation
- Compare lead, lag and lag-lead compensators
- Choose a compensator type for a given deficiency

Syllabus context

Compensators and controllers: introduction to compensators, lag, lead and lag-lead compensators, P, PI and PID controllers.

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Why adjust gain is not enough

Problem Unity feedback $G = K/[s(s + 2)]$. Required $K_v = 20$. What are ζ and M_p with this gain?

- $K_v = K/2 = 20 \Rightarrow K = 40$
- $\omega_n = \sqrt{40} = 6.32$ rad/s; $\zeta = 2/(2 \times 6.32) = 0.158$
- $M_p = \exp(-\pi \times 0.158/\sqrt{(1 - 0.025)}) = 60.5\ \%$

Answer: Accuracy is met but overshoot ≈ 60 %: a compensator is needed

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Compensation

Definition: adding extra components (compensator) to reshape the system so that specifications are met.

Specifications: steady-state accuracy, transient response (Mp, ts), stability margins (GM, PM), bandwidth.

Reasons: gain adjustment alone cannot satisfy conflicting requirements.

Design domains: root locus (s-plane) or Bode (frequency domain).

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Types of compensation

Series (cascade)

Compensator Gc(s) placed in the forward path before the plant. Most common and simplest.

Feedback (parallel)

Compensator in the feedback path, e.g. tachometer (velocity) feedback; reduces effect of plant variations.

Feed-forward

Compensator acts on the reference or disturbance ahead of the loop; improves tracking or rejects measurable disturbances.

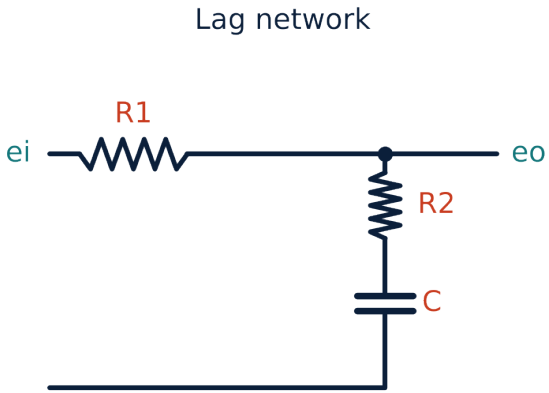
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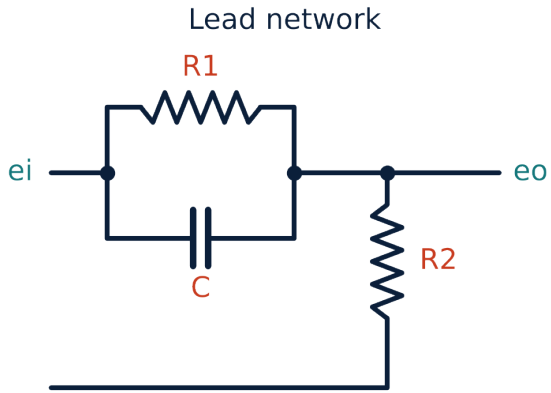
Lead, lag and lag-lead at a glance

Compensator	Form	Effect
Lead	$(1 + aTs)/(1 + Ts)$, $a > 1$	Adds phase lead: ↑ PM, ↑ bandwidth, faster response
Lag	$(1 + Ts)/(1 + \beta Ts)$, $\beta > 1$	Cuts high-frequency gain: ↑ steady-state accuracy, ↓ bandwidth
Lag-lead	Lead × lag	Both: better accuracy and transient
P, PI, PID	Kp, $Kp(1 + 1/Tis)$, …	Controllers: proportional, integral, derivative action

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Lag and lead RC networks





- Lag network: Gc = (1 + sR2C)/[1 + s(R1 + R2)C].
- Lead network: Gc = (1/a)(1 + aTs)/(1 + Ts), a = (R1 + R2)/R2, T = R1R2C/(R1 + R2).

- Both are passive: attenuate at some frequencies.
- Gain loss recovered by an amplifier.

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Key takeaways

- Compensators reshape the loop to meet specifications
- Series, feedback and feed-forward compensation
- Lead: transient; lag: steady-state accuracy; lag-lead: both
- RC networks implement lead and lag

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Practice questions

Q1. Why is gain adjustment alone often inadequate?

Q2. Compare series and feedback compensation.

Q3. Which compensator improves steady-state error? Which improves PM?

Q4. Write the transfer functions of RC lag and lead networks.